

Learning Simultaneous and Sequential Decisions in Multi-Agent Systems With Application to Traffic Signal Control

Haipeng Zhang^{ID}, Zhiwen Wang^{ID}, Jilin Yu, Ruonan Zhang, Caoqing Jiang^{ID}, Gongkun Luo, Weiwei Wu^{ID}, and Wanyuan Wang^{ID}

Abstract—Efficient traffic signal control (TSC) has been one of the most useful ways for reducing urban road congestion. By modeling each intersection as an autonomous agent, multiagent reinforcement learning (MARL) shows remarkable performance in solving dynamic TSC. However, most of TSC methods based on MARL suffer from a non-stationarity problem since agents update their policies simultaneously. To resolve this issue, this paper considers multi-intersection TSC as a multi-agent sequential decision-making process with policy online learning. We utilize a sequential model such as Transformer architecture to learn the multi-agent joint policy. By carefully designing the advantage function of each agent, the monotonic improvement property can be guaranteed. Moreover, to fully exploit the advantages of both simultaneous and sequential MARL, we further propose a novel MARL network selection algorithm (MARL-NS) which selectively employs simultaneous MARL only at states that sequential MARL might fall into local optimum. Our theory proves that MARL-NS preserves cooperative MARL converge properties. Finally, we validate the proposed MARL-NS method on a unified TSC benchmark, LibSignal. Experimental results show that our method can outperform the baseline methods in network-level and arterial coordination.

Index Terms—Sequential decision, switching control, multi-intersection traffic signal control, multi-agent reinforcement learning, Transformer.

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I. INTRODUCTION

THE exponential growth in vehicle volume has led to a pressing societal issue of urban road congestion. One of the most effective ways to reduce traffic congestion is to improve traffic signal control (TSC) systems [1], [2]. It is commonly recognized that modern intelligent TSC should be real-time responsive to dynamic traffic flows and can scale-up to network-level scenarios [3]. To dynamically adjust TSC decisions according to real traffic, reinforcement learning (RL) methods can query the learned policy by mapping real traffic information to the signal decisions [4], [5].

By modeling each intersection as an autonomous agent, multiagent RL (MARL) has emerged as a powerful framework that allows agents to communicate and cooperative with neighbor agents. To enable coordination among agents, existing TSC methods based on MARL have made attempts to reward shaping [6], [7], spatial-temporal influence embedding [8], [9], [10], [11], [12], [13], [14], [15], [16], and network architecture design (e.g., hierarchical architecture [17], [18], [19], and centralized training and decentralized execution (CTDE) framework [20], [21]). Most of these MARL-related TSC methods update agents simultaneously, that is, all agents perform policy improvement at the same time and cannot observe the change of other agents. The simultaneous MARL brings about the non-stationarity problem for network-level TSC with many intersections, i.e., the environment dynamically changes from one agent's perspective as other agents also change their policies [22].

In contrast, the sequential MARL [22], [23] that updates policies agent-by-agent, can alleviate issues of non-stationarity and partial observability of multi-agent systems. This sequential model has demonstrated superior performance compared to state-of-the-art (SOTA) simultaneous MARL techniques in various decision-making systems but has not yet been applied in TSC tasks. Therefore, this paper innovatively formulates network-level TSC as a sequential MARL problem by mapping agents' observation sequences to agents' optimal action sequences. By leveraging the multi-agent advantage decomposition theorem, monotonic performance improvement property can be guaranteed. Nevertheless, sequential MARL suffers from converging to a sub-optimal solution since the pre-determined update order [23]. To tackle

this issue, we introduce a MARL network selection (MARL-NS) algorithm, which optimally selects in the environment to call on sequential MARL to boost simultaneous MARL during training. MARL-NS involves an additional adaptive RL agent that determines at which states to activate sequential MARL while simultaneous MARL is used at all other states.

In summary, the first contribution of this paper is to formulate network-level TSC problem as a sequential MARL task based on Transformer architecture. Second, to fully exploit the advantages of sequential and simultaneous MARL techniques, We propose MARL-NS, which creatively designs an adaptive RL agent that presides over when sequential or simultaneous MARL is used. Finally, we conduct extensive experiments on LibSignal simulation platform to validate the advantages of the proposed algorithm over other existing approaches.

The structure of this paper is as follows: we introduce related TSC approach including simultaneous MARL and sequential MARL methods in Section II. There are relevant traffic terminologies and definitions in Section III. The main details of the proposed algorithm are presented in Section IV. Our experimental settings and comparison result are presented in Section V and a final conclusion is summarized in Section VI.

II. RELATED WORKS

A. Simultaneous MARL for TSC

Effectively coordinated control is necessary for traffic signals at multi-intersection and MARL offers a superior solution. The work of [20] primarily extends the idea of independent Q-learning to advantage actor-critic on multi-intersection TSC without cooperation. In contrast, coordinated control involves communication, graph learning, and reward designing. One study presents TeDA-GCRL as a novel decentralized RL method with a unique graph architecture and time-aware rewards [16]. Additionally, a Transformer-based cooperation mechanism (TCM) [14] features a unique relative position encoding to enhance the representation of traffic conditions. CoSLight [24] employs a top- k collaborator selection policy with a dual-feature extractor. It allows for the effective extraction of phase-level and intersection-level representations while adaptively selecting collaborators with the Transformer for enhanced multi-intersection collaboration. In terms of communication, the study by [25] emphasizes a learned policy for agents to selectively choose message recipients, leading to enhanced congestion management. For scalable learning, SocialLight [26] utilizes a locally centralized critic to estimate individual agent contributions through counterfactual reasoning. X-Light [27] has two types of Transformers with multi-scenario co-training. The lower Transformer aggregates the spatial and temporal information of the road network within a city, and the upper Transformer learns the general decision trajectories across different cities. This design enhances the agent's transferability and enables collaboration across multiple intersections simultaneously. Most methods [10], [11], [15], [19], [28], [29], [30], [31] build complicated network structures, states, and reward functions to

improve the effect of agents' cooperative decisions. However, complicated designs frequently require a lengthy amount of time and a significant number of training samples. Moreover, for simultaneous MARL it is difficult to guarantee convergence and monotonic improvement of the joint policy.

B. Sequential MARL for TSC

With the success of Transformer technique, sequential MARL has been recently developed in TSC. For example, TransformerLight [32] is proposed by formulating TSC control as a sequence modeling problem with the offline supervised learning process described by states, actions, and rewards. DTLight [33] is a Lightweight Decision Transformer-based TSC method that can learn policy from easily accessible offline datasets to achieve offline-to-online. However, existing TSC methods based on sequential MARL mainly focus on single interaction and is offline RL whose evaluation outcomes are restricted by the quality of offline datasets. In contrast, this paper proposes an online sequential MARL method in which each agent accounts for decisions made by preceding agents and can improve their policies by interacting with the TSC environment. Furthermore, we propose the MARL-NS algorithm to learn simultaneous and sequential multi-agent decision-making.

III. PROBLEM FORMULATION

This section formulates the TSC definitions as well as modeling MARL-NS.

A. Traffic Movement

The direction in which vehicles cross from an incoming lane via an intersection to an outgoing lane is referred to as a traffic movement. A traffic movement from lane l to lane m is represented by the symbol (l, m) . As shown in Fig. 1(a), a three-lane (including a right-turn lane) intersection contains 12 traffic movements [4].

B. Signal Phase

A set of permissible traffic movements is known as a traffic signal phase s . S_i represents the set of all phases at the intersection i . According to the phase non-conflict analysis [34], there are eight signal phases at a three-lane intersection, as shown in Fig. 1(b). Phase F is active in Fig. 1(a), and vehicles in the southern left-turn and straight-in lanes are permitted to move to the corresponding outgoing lanes.

C. Pressure of Movement

The distinction in vehicle density between the incoming and the outgoing lanes is defined as the pressure of movement. Where $x(l)$ is the number of vehicles on the lane l and $x_{\max}(l)$ is the maximum number of vehicles allowed on the lane l , the vehicle density of lane l is defined as $x(l)/x_{\max}(l)$. The vehicle capacity of lane m is expressed in

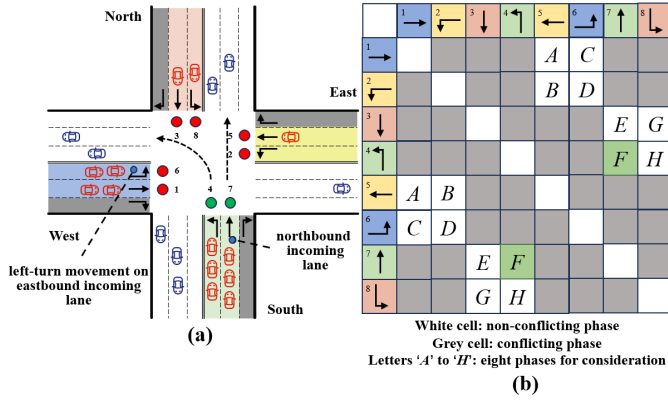


Fig. 1. Definition of traffic movements and traffic signal phases.

the same way. We calculate the pressure $w(l, m)$ of movement (l, m) using Eq. (1).

$$w(l, m) = \frac{x(l)}{x_{\max}(l)} - \frac{x(m)}{x_{\max}(m)} \quad (1)$$

The difference between the number of incoming and outgoing vehicles is shown by $w(l, m)$ if the maximum capacity of each lane $x_{\max}$ is the same.

D. Intersection Pressure

We define a set Z_i consisting of all incoming and outgoing lanes at the intersection i . The pressure P_i at the intersection i is defined as the absolute value of the sum of the pressures in all traffic movements calculated by Eq. (2).

$$P_i = \left| \sum_{l, m \in Z_i} w(l, m) \right| \quad (2)$$

E. Heterogeneous-Agent Mirror Learning and Multi-Agent Transformer

The Heterogeneous-Agent Mirror Learning (HAML) provides a template based on the multi-agent advantage decomposition theorem [35]. The framework can be able to learn to solve multi-agent cooperative tasks. It transforms simultaneous updating of multi-agent policies into sequential updating and optimizes the local objective function. The local objective function of each agent in the updating contains information about the policy updates of the previous agents. The theory has excellent properties, including monotone improvement, value function converges to Nash equilibrium, expected return converges to Nash equilibrium, and solution space consists of Nash equilibrium points properties. The Multi-Agent Transformer (MAT) [23] scheme adopts a Transformer structure with an autoregressive decoder to transform multi-agent sequential updating policies into multi-agent sequential decision-making.

F. Multi-Agent Reinforcement Learning Network Selection for TSC

To tackle the challenges described in TSC, we design a novel MARL-NS algorithm. Our algorithm includes an additional Global RL agent as the switching controller. The Global

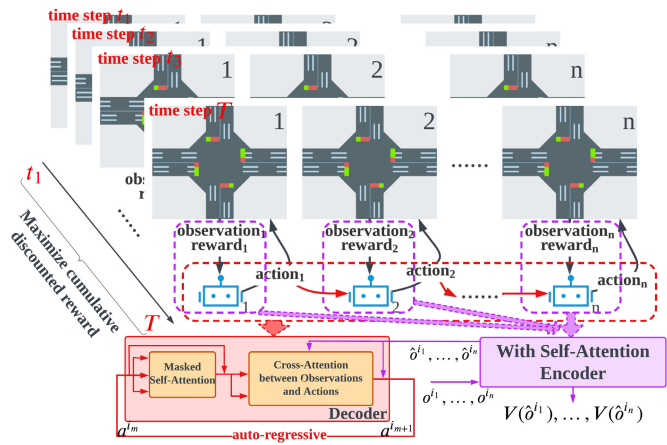


Fig. 2. The sequential decision-making of MARL-NS.

agent selectively employs simultaneous MARL only at states where sequential MARL might fall into local optimum. In the MARL-NS, the components of sequential and simultaneous are independent. The experiences shared are used for training the Global, sequential, and simultaneous components.

IV. OUR METHOD

A. Sequential Modeling TSC

In this part, the multi-intersection traffic signal coordinated control problem is considered as a sequential decision-making process involving multiple agents. To achieve this design, we implement the TSC algorithm based on a MAT model that generates actions autoregressively. Moreover, this approach supports online learning, enabling dynamic policy updates in response to evolving traffic conditions.

The goal is to build a sequential decision-making model M for multi-intersection in order to learn meaningful patterns and generate actions order-by-order with training in online trajectories. The core of sequential multi-intersection TSC framework leverages the Transformer architecture to realize HAML, incorporating both encoder and decoder components. The encoder processes global observations to generate a unified representation, while the decoder generates actions sequentially for each agent, ensuring coordination across multiple agents. Each agent takes into account the decision information of the previous agents when making itself decisions. Fig. 2 illustrates the multi-agent sequential decision-making process within our MARL-NS algorithm.

In the multi-agent sequential decision-making process of MARL-NS, there are n agents to form a permutation $(i_1, \dots, i_n)$. They obtain a joint observation sequence $\mathbf{o} = (o^{i_1}, \dots, o^{i_n})$. The agent's observation o^{i_n} represents the incoming lane's queue length. Then MARL-NS generates a joint action $\mathbf{a}$ by the sequential decision-making scheme. Then the environment feeds back a reward signal R . The reward is defined as the negative values of the pressure at the intersections. Through the continuous training of the model, agents can learn the optimal coordinated control policy of traffic signals at multiple intersections to alleviate traffic congestion. We integrate the Transformer structure with the

actor-critic framework of RL, where the encoder acts as the critic and the decoder acts as the actor.

The encoder, parameterized by ϕ , processes the global observation sequence $\mathbf{o}$ to produce a high-dimensional encoded representation $\hat{\mathbf{o}}^{i:n} = (\hat{o}^{i_1}, \hat{o}^{i_2}, \dots, \hat{o}^{i_n})$. The encoder output $\hat{\mathbf{o}}^{i:n}$ can well reflect the mutual influence of traffic conditions and the high-level interrelationships at different intersections. During the training phase, the encoder minimizes the Bellman error to approximate the value function V_ϕ . The decoder, parameterized by θ , generates the joint action sequence $\mathbf{a}$ in an autoregressive manner. Specifically, it takes the encoded global observation $\hat{\mathbf{o}}^{i:n}$ and previous agents' generated actions $\hat{\mathbf{a}}^{i:m-1}$ as input to predict the next agent's action a^{i_m} . Each autoregressive decoding step involves the following mechanisms.

- **Masked Self-Attention:** It ensures that only preceding actions are considered during the prediction of a^{i_m} , preserving sequential dependencies.
- **Cross-Attention between Observations and Actions:** It models high-level relations between observation features and actions.

The decoder outputs a probability distribution $\pi_\theta^{i_m}(a^{i_m} | \hat{\mathbf{o}}^{i:n}, \hat{\mathbf{a}}^{i:m-1})$, from which actions are sampled. This design allows the decoder to explicitly model dependencies between actions, optimizing joint policies by accomplish multi-agent sequential decision-making. The decoder maximizes the objective function of the Proximal Policy Optimization clip algorithm (PPO-clip). The difference is that the joint advantage function is the sum of the local advantage functions of the sequential decision-making of multiple agents.

In the problem of multi-intersection TSC, the degree of coordination and influence between different intersections are different. For example, a certain intersection only has usually a greater impact on the intersections around it, while the intersections farther away have little impact. Therefore, there is a clear order of influence between the intersections in the entire road network. Through a similar order, the interaction between multiple intersections in the control process can be captured. In addition to the interaction, in order to improve the final effect of traffic signal coordination control, it is necessary to further clarify the contribution of each decision of each intersection to the overall effect in the entire control process.

1) *The Excellent Converge and Monotone Improvement for Sequential:* The sequential decision-making TSC algorithm is abided by the sequential policy-update scheme of the HAML framework. Consequently, it retains all advanced characteristics, including monotone improvement, value function converges to Nash equilibrium, expected return converges to Nash equilibrium, and solution space consisting of Nash equilibrium points properties. The relevant proofs refer to [35].

B. Learning Simultaneous and Sequential

In fact, it is not always necessary to follow the sequential decision-making scheme. So to fully exploit the advantages of simultaneous and sequential multi-agent decision-making,

we can additionally design the Global switching control agent [36]. The Global agent determines when multiple agents make decisions sequentially or simultaneously.

Our MARL-NS introduces a new global RL agent with switching control, which differs from standard RL frameworks. The new adaptive RL agent is trained together with the n cooperative MARL agents in the training phase. The additional Global agent possesses an objective distinct from the n agents resulting in a non-cooperative Markov game $(\mathcal{N} \times \{G\}, \mathcal{S}, (\mathcal{A}, \mathcal{A}_G), (R, R_G), P, \gamma)$.

- $\mathcal{N} = \{1, \dots, n\}$ is the set of MARL agents. G is the Global agent.
- The joint observation space is the product of the local observation spaces of the agents, meaning $\mathcal{S} = \prod_{i=1}^n \mathcal{O}^i$.
- The joint action space is the product of the MARL agents' individual action spaces, meaning $\mathcal{A} = \prod_{i=1}^n \mathcal{A}^i$. $\mathcal{A}_G = \{0, 1\}$ is the Global agent's action set.
- $R : \mathcal{S} \times \mathcal{A} \rightarrow [-R_{\max}, R_{\max}]$ is the joint reward function. $R_G = R - c \cdot \mathbf{1}(g)$ is denoted the Global agent's reward function, where $\mathbf{1}$ is the indicator function and c is cost.
- $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ is the transition probability function.
- $\gamma \in [0, 1)$ is the discount factor.

We prove MARL-NS preserves the MARL convergence properties (in Theorem 1).

1) *MARL-NS's Components:* In the game, the principal components of MARL-NS are composed of a sequential decision-making MARL algorithm, a switching control RL algorithm acting as Global, and a simultaneous decision-making MARL algorithm. Our algorithm is illustrated in Fig. 3.

- **Full sequential decision-making.** This component combines the Transformer and PPO algorithms to realize an on-policy online learning method of sequential multi-intersection TSC. See Section IV-A for details. We consider that the symbol π^s is the sequential decision-making policy.
- **Full simultaneous decision-making.** For simultaneous decision-making, we designed and delivered a version of CTDE rather than [23]. We replace the autoregressive decoder with an encoder rather than MLP. The encoder is shared among multiple agents and its input is the agent's observations. We consider that the symbol π^d is the simultaneous decision-making policy.
- **Switching control (Global).** The Global policy's action set $\mathcal{A}_G = \{0, 1\}$ consists of 2 actions: 1) use the sequential decision policy, 2) use the simultaneous decision policy. The Global updates its policy g .

In the MARL-NS, the Global agent observes the global state s_t of the environment and samples the discrete policy of the switching controller $g_t \sim g : \mathcal{S} \rightarrow \{0, 1\}$. The DQN module of the Global agent receives input that concatenates the global state of the environment with the policy actions from the simultaneous decision scheme and the sequential decision scheme. Then we get g_t by Eq. (3) with the switching policy g of Global agent with the outputs of DQN. If $g_t = 1$, we select the sequential decision-making to get the joint actions. If $g_t = 0$, $\mathbf{a}_t$ is

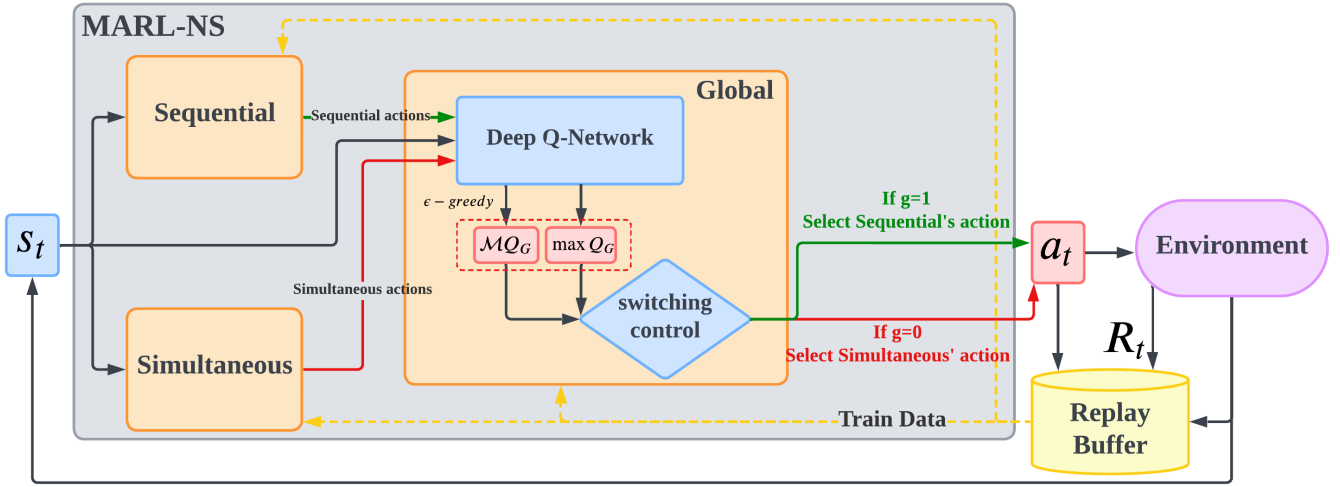


Fig. 3. The architecture of our MARL-NS algorithm. It comprises Sequential, Simultaneous decision-making, and a Global switching control agent. The Global agent, including a DQN, estimates two Q-values (MQ_G and $\max Q_G$) and uses a binary decision g to select a decision-making mode. The chosen action a_t is executed in the environment, and the samples of the current episode are used to update the algorithm with the global loss function L_G and PPO-clip objective function and discarded after updating. This architecture enables adaptive decision-making between sequential and simultaneous decision-making modes, optimizing performance in MARL scenarios.

generated from the simultaneous. The difference is that the simultaneous decision-making scheme is always the maximization of Q-value, while the sequential decision-making scheme follows the ϵ -greedy policy. The terms of $\max Q_G$ and MQ_G respectively express the Q-value of simultaneous decision scheme and the Q-value of the sequential decision selection under the intervention of Global agent. These two items are the estimated value for the global switch control agent's DQN to obtain based on the input. In our newly extended loss function by Eq. (5), the new operator will get the target value of DQN to train the Global agent. In particular, there is independence between the policies of sequential and simultaneous. The experiences generated by the process are used to train the Global, sequential, and simultaneous parts and discarded after every training. The Global agent achieves its objective of improving the learning process and maximizing the performance of the system of the n MARL agents through its decisions on where to utilize Sequential decision-making. Frequently using a Transformer structure with autoregression to accomplish multi-agent decision-making order by order on a basis may result in communication costs and delays because most autonomous agents typically exist in a distributed manner in the real world. To induce Global to selectively choose to undertake sequential decision-making rather than full sequential, each utilization of sequential incurs a fixed cost for Global. The cost is measured by a constant $c > 0$.

For the training of the entire system, we need to define the learning method of the Global agent.

2) *The Global Agent's Policy:* For any $s_t \in \mathcal{S}$ and all $a_t \in \mathcal{A}$, the policy g of Global agent is given by Eq. (3).

$$g(\cdot | s_t) = \mathbf{1} \left(\mathcal{M}^{g, \pi^s} Q_G(s_t, a_t) - \max_{a_t \in \mathcal{A}} Q_G(s_t, a_t | \pi^d, g) \right) \quad (3)$$

It provides the characterization of where Global should perform Sequential decision-making. During the learning phase,

the condition allows the characterization to be evaluated online. The term of $\mathcal{M}^{g, \pi^s} Q_G(s_t, a_t) = Q_G(s_t, \pi^s(s_t) | \cdot) - c$ measures the expected return for Global after switching to the sequential decision-making joint policy minus the intervention cost c . $\mathbf{1}$ is the indicator function. It is defined as $\mathbf{1}(x) = 1$ if $x > 0$, and $\mathbf{1}(x) = 0$ otherwise.

Definition 1: We extend the following Q-learning variant:

$$Q_t(s_t, a_t) = Q_t(s_t, a_t) + \alpha \left[\max \left\{ \mathcal{M}^{g, \pi^s} Q_t(s_t, a_t), R_G + \gamma \sup_{a' \in \mathcal{A}} Q_t(s', a') \right\} - Q_t(s_t, a_t) \right]. \quad (4)$$

α is the learning rate. We use a neural network to approximate the result.

Therefore, we employ the extended Deep Q-Network of definition 1 to train the Global agent of MARL-NS. Specifically, the Global agent minimizes the loss function by Eq. (5).

$$L_G = \frac{1}{2T} \sum_{t=0}^{T-1} \left[Q_G(s_t, a_t) - \max \left\{ \mathcal{M}^{g, \pi^s} Q_G(s_t, a_t), R_G + \gamma \max_{a' \in \mathcal{A}} Q_G(s', a') \right\} \right]^2 \quad (5)$$

The whole process of MARL-NS is as Algorithm 1 in part B of supplementary material.

C. Convergence and Optimality of MARL-NS

For a set of joint sequential and simultaneous policies, the solution of Global's problem is a limit point of a sequence of Bellman operations acting on a value function in theorem 1. The proof is in part A of the Appendix.

Theorem 1: Let $v_G : \mathcal{S} \rightarrow \mathbb{R}$, for any policies π^s , the solution of Global's problem is provided by Eq. (6).

$$\lim_{k \rightarrow \infty} T_G^k v_G(\cdot | \pi, \mathbf{g}) = \max_{\hat{\mathbf{g}}} v_G(\cdot | \pi, \hat{\mathbf{g}}) \quad (6)$$

T_G is a contraction, and the Bellman operator is $T_G v_G = \max \{ \mathcal{M}^{\mathbf{g}, \pi^s} Q_G, \max_{a \in \mathcal{A}} [R_G + \gamma \sum_{s' \in \mathcal{S}} P(s' | \cdot) v_G(s')] \}$. Given a system of convergent MARL learners, it ensures the convergence of the system when Global uses a Q-learning variant. Then we can obtain the performance outcomes. There exists some finite integer N such that $v(s | \tilde{\pi}_m) \geq v(s | \pi_m)$, $\forall s \in \mathcal{S}$ for any $m \leq N$. $\tilde{\pi}_m$ and π_m are the joint policies after the m^{th} learning iteration with and without Global's influence respectively.

The theorem 1 shows that the system in which both the Sequential decision-making, Simultaneous decision-making, and Global switching control agents train concurrently converges to a solution. Using the MARL-NS framework leads to improvements in the underlying MARL systems.

V. EXPERIMENTS

We experimentally verify the control effect and convergence of the proposed algorithm on network-level and arterial datasets. All RL approaches are learned online without any pre-training process on the same few-shot space. All experiments are conducted on the LibSignal [37] platform that includes the CityFlow [38] simulator. The experiment of each RL algorithm on each dataset is repeated 5 times and all results are averaged.

A. Description of Datasets

The setting follows each green phase with a three-second yellow and a two-second all-red phase, as is customary. In the network-level experiments, we use the real datasets of the Hangzhou and Jinan regional road networks.¹ The road networks and traffic flows of datasets refer to [8]. Since the two datasets differ in road network structure and traffic flow characteristics, the generalization of our algorithm can be verified to a certain extent.

Synthetic traffic flow data from the 6×1 road are used in arterial experiments. In order to verify the adaptability of the algorithm to different traffic flows, we selected two kinds of data, flat and peak.² The road networks and traffic flows of datasets refer to [6].

B. Evaluation Metrics

We utilize average traveling time (ATT) and average intersection pressure (AIP) to evaluate the performance of various TSC models. The total time (in seconds) it takes for all vehicles to arrive and exit the region is averaged to calculate the average traveling time. It is the most common performance metric for TSC problems in the traffic domain. The pressure represents the degree of imbalance in vehicle density between

TABLE I
THE INTERSECTION WITH THE HIGHEST TRAFFIC FLOW
(VEHICLES/HOUR)

item	Hangzhou _{4×4}	Jinan _{3×4}	Synthetic _{6×1}	
			Flat	Peak
ID	1_4	1_1	2_1	5_1
flow	1236	1428	1321	1306

the entering and exiting lanes. Minimizing the pressure can effectively be utilized to optimize the throughput.

In order to prove that MARL-NS has coordinated control, we choose the intersections with the largest traffic flow in the road network to observe the impact of MARL-NS on the traffic conditions at a specific intersection. The outcomes of the selection are that Hangzhou is 1_4, Jinan is 1_1, Flat is 2_1, and Peak is 5_1. For example, ID 1_4 is defined as the intersection represented by the first column from west to east and the fourth row from south to north in the road network. The results of the selection are revealed in TABLE I.

C. Contrast Method

Our method is compared with the baseline methods listed below, which include both traditional and RL methods. The action interval is set to 10 seconds. Each episode is comprised of a 60-minute simulation.

- FixedTime [7]: Fixed-time control is a control method in which the intersection runs continuously and repeatedly according to a preset fixed timing scheme.
- MaxPressure [39]: Max-pressure control greedily chooses the phase that maximizes pressure based on upstream and downstream queue lengths.
- SOTL [7]: Self-Organizing Traffic Light Control involves manually adjusting a threshold for the number of waiting vehicles. When the number of waiting vehicles exceeds the manually adjusted threshold, the traffic signal will be changed.
- PressLight [6]: PressLight is a DQN-based RL approach that combines the concept of pressure to design state and reward. The state includes the number of vehicles on each incoming lane, the current phase s , and the number of vehicles on each outgoing lane.
- CoLight [8]: CoLight is a DQN-based RL method using Graph Attention RL. It optimizes the queue length using information from neighbors, and the queue length is employed by the reward function.
- MPLight [4]: MPLight is a DQN-based RL method using parameter sharing. It utilizes RL approaches and transportation theories to address the problem of multi-intersection TSC, particularly for large-scale networks. Phase pressure and current phase s are used as the observation.

D. Performance Comparison

We set the memory capacity of the replay buffer for PressLight, CoLight, and MPLight to 360. It is the same

¹https://github.com/traffic-signal-control/RL_signals

²https://github.com/DaRL-LibSignal/LibSignal/tree/master/data/raw_data

TABLE II
SAMPLE EFFICIENCY FOR DIFFERENT METHODS

Models	replay buffer size (number of samples)	number of samples used per update	learning way	update number (epoches/episode)	update frequency (number of interval decisions)	whether to clear replay buffer after updating
CoLight	360	randomly select 16 samples from replay buffer	off-policy	360	1 (Temporal-Difference update)	False
MPLight	360	randomly select 16 samples from replay buffer	off-policy	360	1 (Temporal-Difference update)	False
PressLight	360	randomly select 16 samples from replay buffer	off-policy	360	1 (Temporal-Difference update)	False
MARL-NS	360	a complete trajectory of each episode	on-policy	15	360 (Monte-Carlo update)	True

TABLE III
EVALUATIONS OF DIFFERENT METHODS

Models	Average Traveling Time(s)				Average Intersection Pressure			
	Hangzhou _{4×4}	Jinan _{3×4}	Flat _{6×1}	Peak _{6×1}	Hangzhou _{4×4}	Jinan _{3×4}	Flat _{6×1}	Peak _{6×1}
FixedTime	575.56	425.31	149.36	162.51	173.23	83.41	53.19	64.65
MaxPressure	365.06	328.96	99.41	99.66	24.39	34.99	20.80	20.14
SOTL	354.13	333.58	117.22	120.40	32.09	33.63	34.77	36.37
CoLight	343.54	682.42	98.20	98.73	23.01	271.17	18.92	20.11
MPLight	334.54	311.21	96.41	98.05	20.48	28.20	18.72	19.30
PressLight	436.81	430.16	110.92	106.66	43.93	78.02	29.69	25.65
MARL-NS	318.40	282.89	93.36	95.34	18.65	17.87	16.57	18.13

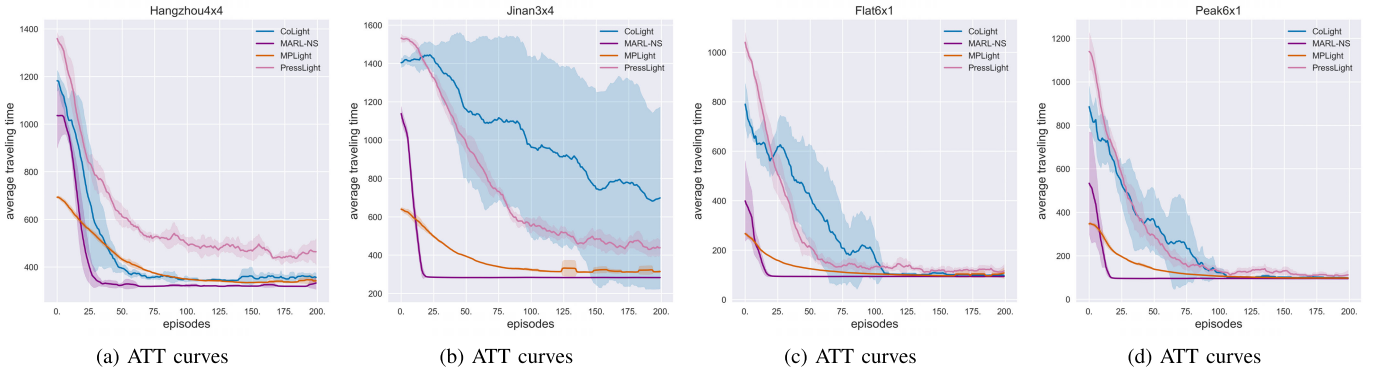


Fig. 4. ATT of different methods.

number of decisions made by the simulation in an hour. MARL-NS has only one current full 60-minute trajectory for each policy update. This makes RL models learn in a few-shot space. Moreover, MARL-NS is a Monte-Carlo update for 15 epochs, while CoLight, MPLight, and PressLight are Temporal-Difference updates. In this basically identical sample space, MARL-NS's sample efficiency is higher than that of CoLight, MPLight, and PressLight. The MARL-NS exhibits characteristics of on-policy learning. The sample collection is performed using the policy in effect during the current episode, and after every training, these samples are discarded. In the MARL-NS, the global switching control policy part, the sequential decision-making part and the simultaneous decision-making part are only trained on the current episode's data without reusing older samples. Given these features, our algorithm can be described as an on-policy. The related settings are listed in TABLE II.

The shaded part in the graph is the standard error interval, which represents the stability of the algorithm. For traditional methods like FixedTime, MaxPressure, and SOTL, there is no learning process, so there are no relevant curves in the experiments. MARL-NS is compared

against baseline methods on several publicly available datasets. The findings of the experiment indicate that our algorithm works better on these publicly available datasets than baseline methods.

1) *Metric Analysis:* In the TABLE III, we present a comparison of MARL-NS with both traditional traffic control methods (FixedTime, MaxPressure, SOTL) and RL-based baselines (CoLight, MPLight, PressLight) across various traffic datasets. This comparison encompasses both network-level and arterial coordination metrics. The results demonstrate that MARL-NS consistently outperforms all other methods in terms of ATT and AIP across all the datasets.

2) *Convergence Analysis:* In the Fig. 4(a) and (b), the ATT curves are depicted over increasing episodes. On both the Hangzhou and Jinan datasets, MARL-NS demonstrates superior convergence speed and stability compared to others. The corresponding AIP curves in Fig. 5(a) and (b) show effective vehicle distribution, with MARL-NS's trends aligning with its ATT performance, validating the use of intersection pressure as an efficient TSC optimization metric. Additionally, Fig. 4(c) and (d) reveal that MARL-NS significantly exhibits faster and more stable convergence in terms of ATT on both

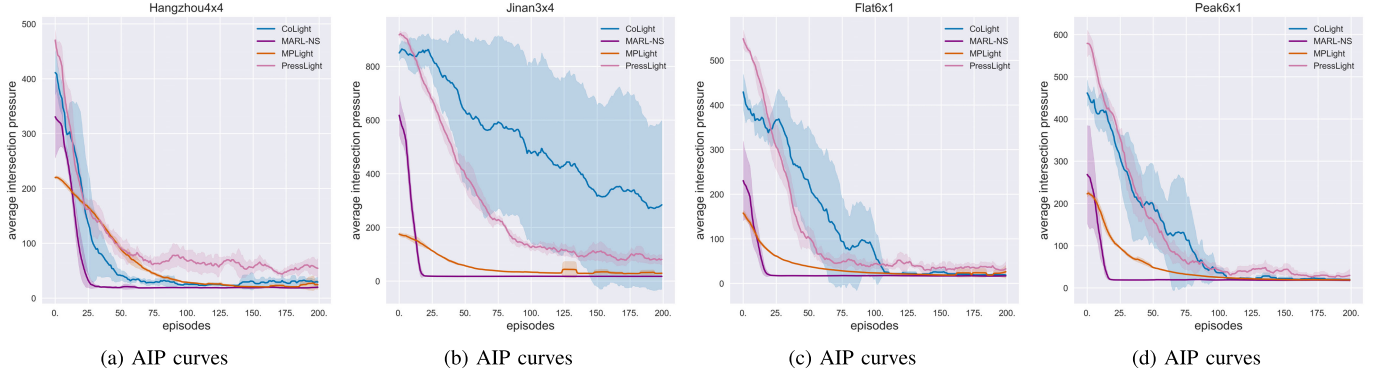


Fig. 5. AIP of different methods.

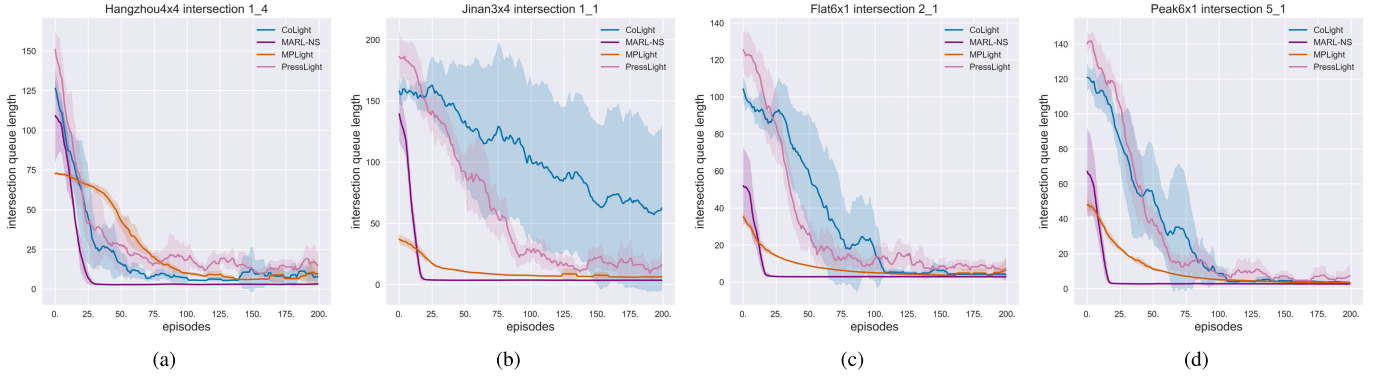


Fig. 6. Single-intersection queue length of different methods.

TABLE IV
QUEUE LENGTH OF SINGLE-INTERSECTION

Models	Single-intersection Queue Length			
	Hangzhou _{4×4}	Jinan _{3×4}	Flat _{6×1}	Peak _{6×1}
CoLight	5.39	57.35	4.00	3.59
MPLight	5.98	6.02	3.59	3.53
PressLight	9.55	9.28	6.50	4.44
MARL-NS	2.76	3.52	2.85	2.67

TABLE V
EVALUATIONS OF ABLATION STUDIES

Models	Average Traveling Time(s)			
	Hangzhou _{4×4}	Jinan _{3×4}	Flat _{6×1}	Peak _{6×1}
Sequential	322.08	283.75	93.89	97.05
Simultaneous	378.46	289.94	94.67	96.93
MARL-NS	318.40	282.89	93.36	95.34

flat and peak flows. This trend of superior performance in AIP by MARL-NS is further confirmed in flat and peak traffic conditions, as illustrated in Fig. 5(c) and (d).

3) *Coordination Analysis*: The TABLE IV presents the queue length values of MPLight, CoLight, PressLight, and MARL-NS at a single intersection, with MARL-NS showing the best results. The relevant graphs are in Fig. 6. Fig. 6(a) illustrates that MARL-NS achieves the fastest convergence at Hangzhou's 1_4 intersection. At Jinan's 1_1 intersection, as shown in Fig. 6(b), MARL-NS demonstrates superior convergence in both speed and stability compared to the notably unstable CoLight, PressLight, and slower MPLight. Furthermore, Fig. 6(c) and (d) show that MARL-NS maintains lower and more stable queue lengths at intersections 2_1 and 5_1 for both flow conditions. The combined analysis of Fig. 4 and Fig. 5 indicates effective coordinated control by our model.

E. Ablation Studies

To explore the effectiveness and necessity of the global switching control, we conduct some ablation studies. For this purpose, we implement two algorithms: completely sequential decision-making and completely simultaneous decision-making. As demonstrated in TABLE V, MARL-NS outperforms sequential and simultaneous methods in terms of ATT results. In terms of convergence and stability, MARL-NS outperforms sequential and simultaneous, as demonstrated in Fig. 7. It illustrates that the system as a whole receives an improved control effect after global switching control is added. The relevant graphs are in Fig. 7.

Meanwhile, we also mainly explore the trend regarding the selection of sequential method in MARL-NS, as shown in Fig. 8. This figure illustrates the counts of sequential decision-making in our algorithm over episodes. The environments tested include Hangzhou, Jinan, and arterial. The y-axis represents the number of sequential decision-making counts, while the x-axis indicates the progression of training episodes.

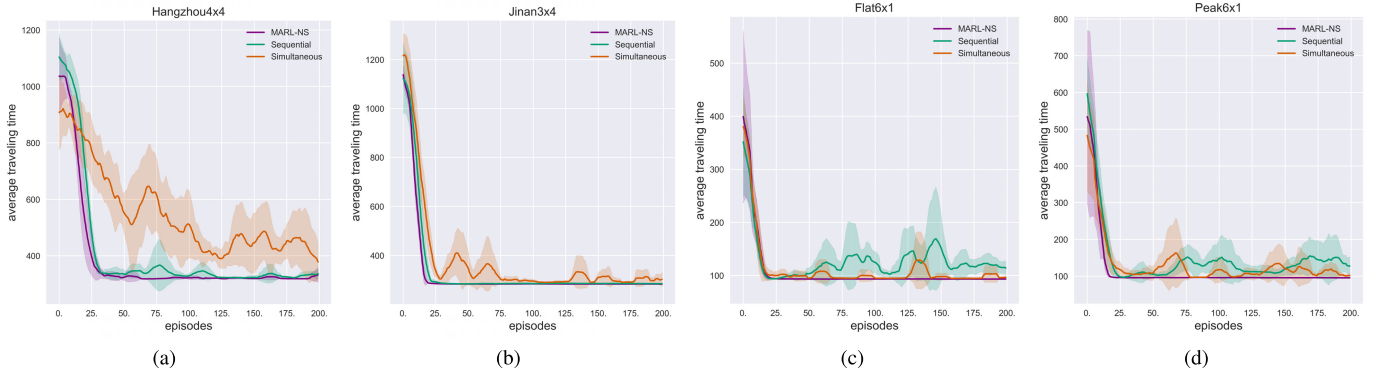


Fig. 7. Evaluations ATT of ablation studies.

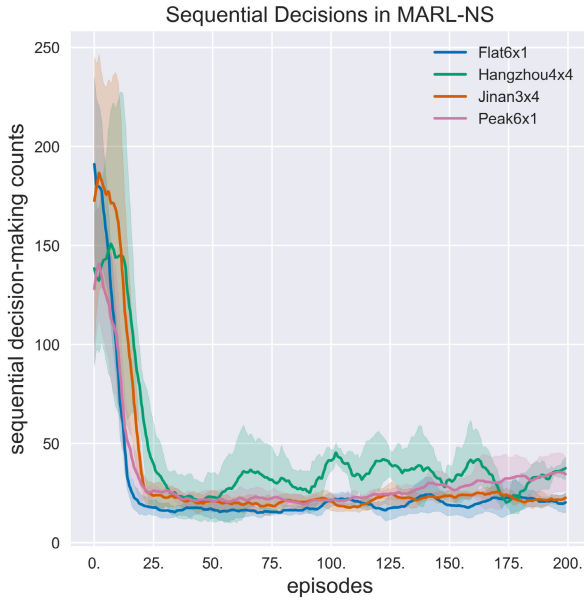


Fig. 8. Evaluations the sequential decision-making counts of MARL-NS.

In the initial episodes (up to approximately episode 20), all environments display high sequential decision-making counts, suggesting that the agents are in the exploratory phase and use the sequential decision-making scheme to improve learning. By around episode 50, most environments demonstrate stabilized sequential decision-making counts with minor fluctuations, implying that the agents have largely optimized their policies and require fewer sequential decisions according to environment conditions to operate effectively. Notably, the Hangzhou dataset shows some variability and a slight increase in decision counts after episode 100. This may indicate environmental challenges or dynamic shifts that disrupt the learned policies, necessitating occasional adjustments from the agents. For the case, our MARL-NS selectively uses sequential decision-making and simultaneous decision-making to be stable in TSC.

VI. CONCLUSION AND FUTURE WORK

This paper studies the problem of non-stationarity in TSC. The objective is to maximize assignments' average

traveling time. We propose a novel MARL network selection algorithm as MARL-NS, which can learn simultaneous and sequential decision-making selectively in multi-agent systems of TSC. In the MARL-NS, the sequential decision component involves Transformer, PPO, and Heterogeneous-Agent Mirror Learning's sequential update scheme combined. Meanwhile, the Global agent makes switching control and RL combining. Our algorithm can learn online and preserve converge properties. Experimental results show that on both network-level and arterial datasets, the proposed algorithm achieves higher performance in few-shot space than other benchmark algorithms. In the meantime, we discover the global Q-value function takes the joint actions from all agents as input, which may raise potential concerns on scalability issues. We will continually improve the algorithm in the future.

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